

ACTUF502 Assurance Vie II

Prof. Jennifer Alonso García

Stochastic Mortality Modeling: Mortality forecasting in R with StMoMo package

Based on the explanatory slides of dr. Andrés Villegas (UNSW Sydney, CEPAR).

Agenda

- Recap of Stochastic Mortality Models
- Mortality modelling in R
- Generalised Age-Period-Cohort mortality models
- **StMoMo** package

Advances in mortality modelling

- **Lee-Carter model** (Lee and Carter, 1992)
 - ▷ Add more bilinear age-period components (Renshaw and Haberman, 2003)
 - ▷ Add a cohort effect (Renshaw and Haberman, 2006)
- Two factor **CBD model** (Cairns et al., 2006)
 - ▷ Add cohort effect, quadratic age term (Cairns et al., 2009)
 - ▷ Combine with features of the Lee-Carter (Plat, 2009)
- **Many more models** proposed in the literature (e.g. Aro and Pennanen (2011); O'Hare and Li (2012); Börger et al. (2014); Alai and Sherris (2014))

Mortality modelling in R

- **Demography** (Hyndman et al., 2011)
 - ▷ Function for downloading data from the Human Mortality Database
 - ▷ Lee-Carter model and several of its variants
- **ilc** (Butt et al., 2014)
 - ▷ Lee-Carter with cohorts and Lee-Carter under a Poisson framework
- **Lifemetrics**
(<http://www.macs.hw.ac.uk/~andrewc/lifemetrics/>)
 - ▷ CBD and extensions
 - ▷ Lee-Carter with cohorts and Lee-Carter under a Poisson framework

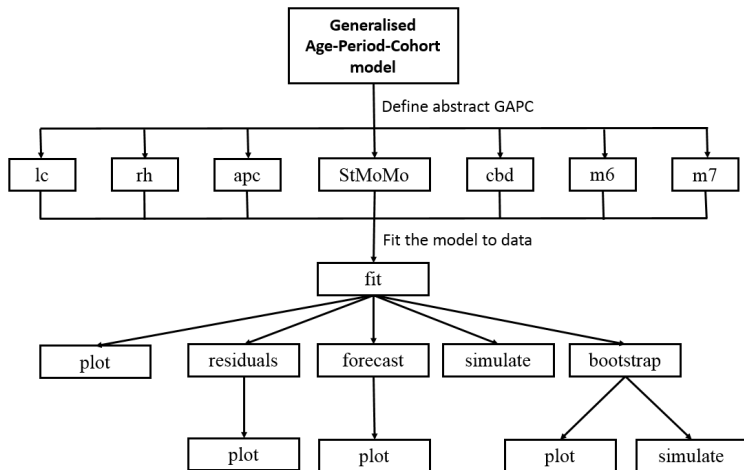
Limitation of existing R packages

- Not easily expandable to include new models
- Limited forecasting and simulation capabilities
- Limited tools for goodness-of-fit analysis
- Do not allow for parameter uncertainty
- **StMoMo** seeks to overcome these limitations

StMoMo: An R package for **St**ochastic **M**ortality **M**odelling

- On CRAN:
`http://cran.r-project.org/web/packages/StMoMo/`
- Development version on Github:
`https://github.com/amvillegas/StMoMo`
- the stable version on R CRAN:
`install.packages("StMoMo")`
- To load the package within R:
`library(StMoMo)`

Overview of the structure of **StMoMo**: Installation and loading



Generalised Age-Period-Cohort stochastic mortality models

StMoMo is based on the unifying framework of the family of Generalised Age-Period-Cohort stochastic mortality models

- General Age-Period-Cohort model structure (Hunt and Blake, 2018)
- Generalised (non-)linear model (Currie, 2016)

Generalised Age-Period-Cohort stochastic mortality models

1. Random Component:

$$D_{xt} \sim \text{Poisson}(E_{xt}^c \mu_{xt}) \quad \text{or} \quad D_{xt} \sim \text{Binomial}(E_{xt}^0, q_{xt})$$

2. Systematic Component:

$$\eta_{xt} = \alpha_x + \sum_{i=1}^N \beta_x^{(i)} \kappa_t^{(i)} + \beta_x^{(0)} \gamma_{t-x}$$

▷ Lee-Carter type $\rightsquigarrow \beta_x^{(i)}$, non-parametric

▷ CBD type $\rightsquigarrow \beta_x^{(i)} \equiv f^{(i)}(x)$, pre-specified parametric function

3. Link Function:

$$g\left(\mathbb{E}\left(\frac{D_{xt}}{E_{xt}}\right)\right) = \eta_{xt}$$

▷ log-Poisson: $\eta_{xt} = \log \mu_{xt}$

▷ logit-Binomial: $\eta_{xt} = \text{logit } q_{xt}$

Generalised Age-Period-Cohort stochastic mortality models

4. Set of parameter constraints:

- ▷ Most mortality models are only identifiable up to a transformation
- ▷ Need parameters constraints to ensure identifiability
- ▷ **Constraint function** v mapping an arbitrary vector of parameters

$$\theta := \left(\alpha_x, \beta_x^{(1)}, \dots, \beta_x^{(N)}, \kappa_t^{(1)}, \dots, \kappa_t^{(N)}, \beta_x^{(0)}, \gamma_{t-x} \right)$$

into a vector of transformed parameters

$$v(\theta) = \tilde{\theta} = \left(\tilde{\alpha}_x, \tilde{\beta}_x^{(1)}, \dots, \tilde{\beta}_x^{(N)}, \tilde{\kappa}_t^{(1)}, \dots, \tilde{\kappa}_t^{(N)}, \tilde{\beta}_x^{(0)}, \tilde{\gamma}_{t-x} \right)$$

satisfying the model constraints with no effect on the predictor η_{xt} (i.e. θ and $\tilde{\theta}$ result in the same η_{xt})

GAPC stochastic mortality models with **StMoMo**

GAPC model are constructed using the function

`StMoMo(link, staticAgeFun, periodAgeFun, cohortAgeFun, constFun)`

- `link`: defines the **link** and **random component**.
- The **predictor** is defined via:
 - ▷ `staticAgeFun`: logical indicating if α_x is present.
 - ▷ `periodAgeFun`: list of length N defining $\beta_x^{(i)}$, $i = 1, \dots, N$.
 - ▷ `cohortAgeFun`: defines parameter $\beta_x^{(0)}$
- `constFun`: Implementation of **constraint function** $v(\theta) = \tilde{\theta}$ which defines the **set of parameter constraints**

$$\eta_{xt} = \alpha_x + \sum_{i=1}^N \beta_x^{(i)} \kappa_t^{(i)} + \beta_x^{(0)} \gamma_{t-x}$$

GAPC stochastic mortality models with **StMoMo**

Model	Predictor (η_{xt})
LC	$\alpha_x + \beta_x^{(1)} \kappa_t^{(1)}$
CBD	$\kappa_t^{(1)} + (x - \bar{x}) \kappa_t^{(2)}$
APC	$\alpha_x + \kappa_t^{(1)} + \gamma_{t-x}$
M7	$\kappa_t^{(1)} + (x - \bar{x}) \kappa_t^{(2)} + ((x - \bar{x})^2 - \hat{\sigma}_x^2) \kappa_t^{(3)} + \gamma_{t-x}$

- For consistency, all under a log-Poisson setting:

$$D_{xt} \sim \text{Poisson}(E_{xt}^c \mu_{xt})$$

$$\log \mu_{xt} = \eta_{xt}$$

Lee-Carter model (Lee and Carter, 1992)

Predictor: $\eta_{xt} = \alpha_x + \beta_x^{(1)} \kappa_t^{(1)}$

Constraints: $\sum_x \beta_x^{(1)} = 1, \quad \sum_t \kappa_t^{(1)} = 0$

$$v(\theta) = \tilde{\theta}: \left(\alpha_x, \beta_x^{(1)}, \kappa_t^{(1)} \right) \rightarrow \left(\alpha_x + c_1 \beta_x^{(1)}, \frac{1}{c_2} \beta_x^{(1)}, c_2 (\kappa_t^{(1)} - c_1) \right)$$

$$\text{with } c_1 = \frac{1}{n} \sum_t \kappa_t^{(1)} \quad c_2 = \sum_x \beta_x^{(1)}$$

#Define constraint function

```
constLC <- function(ax, bx, kt, b0x, gc, wxt, ages){  
  c1 <- mean(kt[1, ], na.rm = TRUE)  
  c2 <- sum(bx[, 1], na.rm = TRUE)  
  list(ax = ax + c1 * bx[, 1], bx[, 1] = bx[, 1] / c2,  
        kt[1,] = c2 * (kt[1, ] - c1))}
```

#Define model

```
LC <- StMoMo(link = "log", staticAgeFun = TRUE,  
             periodAgeFun = "NP", constFun = constLC)
```

CBD model (Cairns et al., 2006)

Predictor: $\eta_{xt} = \kappa_t^{(1)} + (x - \bar{x})\kappa_t^{(2)}$

Constraints: No constraints necessary

```
#B2: x - \bar{x}
f2 <- function(x, ages) x - mean(ages)
#Define model
CBD <- StMoMo(link = "log", staticAgeFun = FALSE,
periodAgeFun = c("1", f2))
```

Model definition: Predefined functions for common models

```
LC <- lc()
CBD <- cbd(link = "log")
APC <- apc()
M7 <- m7(link = "log")
```

```
## Poisson model with predictor:
log m[x,t] = a[x] + b1[x] k1[t]
## Poisson model with predictor:
log m[x,t] = k1[t] + f2[x] k2[t]
## Poisson model with predictor:
log m[x,t] = a[x] + k1[t] + g[t-x]
## Poisson model with predictor:
log m[x,t] = k1[t] + f2[x] k2[t] +
f3[x] k3[t] + g[t-x]
```


Model fitting: Data

Sample data for England and Wales males aged 0-100 for the period 1961-2011

```
Dxt <- EWMaleData$Dxt
Ext <- EWMaleData$Ext
ages <- EWMaleData$ages    #0-100
years <- EWMaleData$years  #1961-2011
```

Dxt

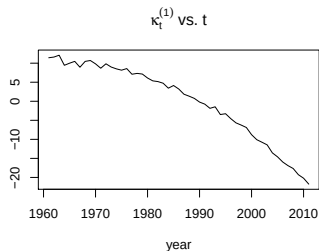
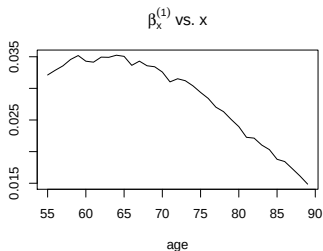
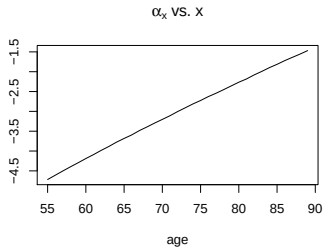
##	1961	1962	1963	1964	1965	1966	1967	1968	1969
## 0	9988	10573	10401	10011	9518	9357	8673	8705	8331
## 1	665	598	665	588	571	616	549	552	567
## 2	398	353	378	354	354	389	374	381	381
## 3	249	259	261	254	292	301	281	316	275

Model fitting

```
#Ages for fitting
ages.fit <- 55:89
#Fit other models
LCfit <- fit(LC, Dxt = Dxt, Ext = Ext, ages = ages,
years = years, ages.fit = ages.fit)
APCfit <- fit(APC, Dxt = Dxt, Ext = Ext, ages = ages,
years = years, ages.fit = ages.fit)
CBDfit <- fit(CBD, Dxt = Dxt, Ext = Ext, ages = ages,
years = years, ages.fit = ages.fit)
M7fit <- fit(M7, Dxt = Dxt, Ext = Ext, ages = ages,
years = years, ages.fit = ages.fit)
```

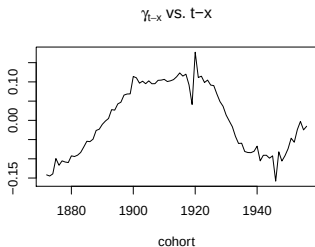
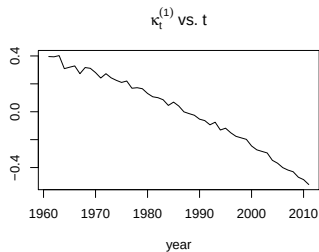
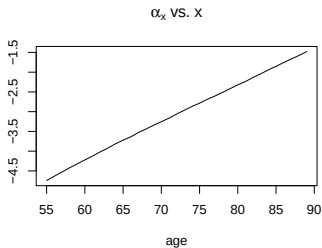
Parameter estimates: Lee-Carter

`plot(LCfit)`



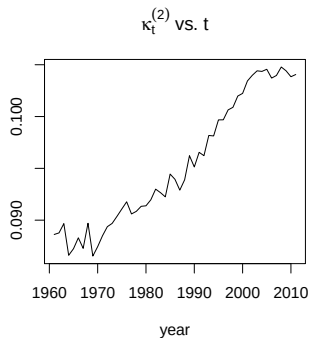
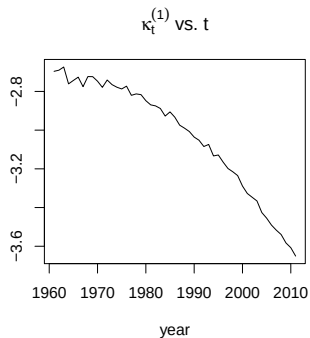
Parameter estimates: APC

```
plot(APCfit, parametricbx = FALSE)
```



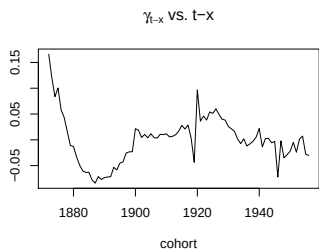
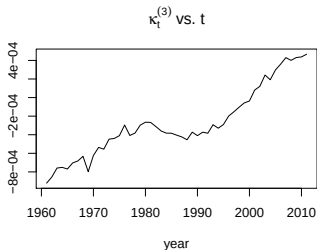
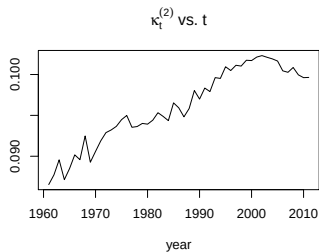
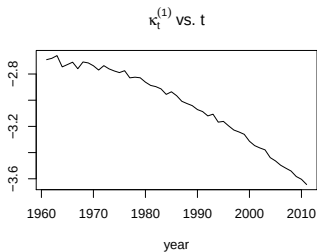
Parameter estimates: CBD

```
plot(CBDfit, parametricbx = FALSE)
```



Parameter estimates: M7

```
plot(M7fit, parametricbx = FALSE)
```



Goodness-of-fit: Residuals

Examine deviance residuals of the model

$$r_{xt} = \text{sign}(d_{xt} - \hat{d}_{xt}) \sqrt{\frac{\text{dev}(x, t)}{\hat{\phi}}}$$

- Patterns $\rightsquigarrow$ not all features of the data captured

#Compute residuals

```
LCres <- residuals(LCfit)
```

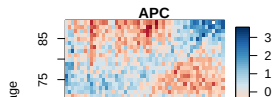
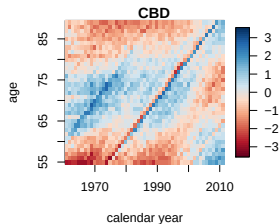
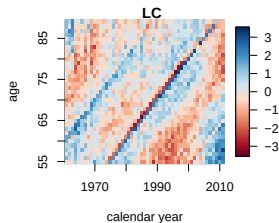
```
CBDres <- residuals(CBDfit)
```

```
APCres <- residuals(APCfit)
```

```
M7res <- residuals(M7fit)
```

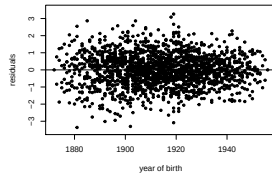
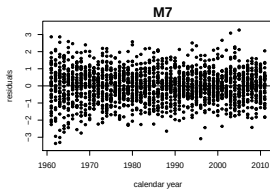
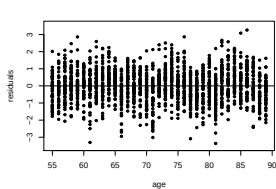
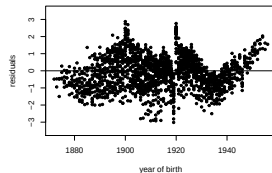
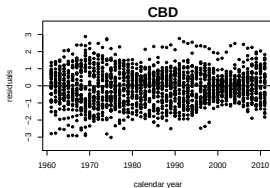
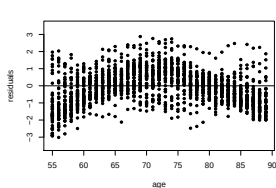
Goodness-of-fit: Residual heatmaps

```
plot(LCres, type = "colourmap", reslim = c(-3.5, 3.5))
```



Goodness-of-fit: Residual Scatter-plots

```
plot(CBDres, type = "scatter", reslim = c(-3.5, 3.5))  
plot(M7res, type = "scatter", reslim = c(-3.5, 3.5))
```



Goodness-of-fit vs Parsimony: AIC and BIC

$$AIC = -2\mathcal{L} + 2 \times \nu \qquad BIC = -2\mathcal{L} + \log K \times \nu$$

#Compute residuals

AIC(CBDfit)

[1] 40374.87

BIC(CBDfit)

[1] 40934.56

	LC	APC	CBD	M7
AIC	30566	25344	40375	21721
BIC	31219	26266	40935	23010

Forecasting and simulation

- **Period indexes:** Multivariate random walk with drift

$$\boldsymbol{\kappa}_t = \boldsymbol{\delta} + \boldsymbol{\kappa}_{t-1} + \boldsymbol{\xi}_t^{\kappa}, \quad \boldsymbol{\kappa}_t = \begin{pmatrix} \kappa_t^{(1)} \\ \vdots \\ \kappa_t^{(N)} \end{pmatrix}, \quad \boldsymbol{\xi}_t^{\kappa} \sim N(\mathbf{0}, \Sigma),$$

- **Cohort effect:** ARIMA(p, q, d) with drift

$$\Delta^d \gamma_c = \delta_0 + \phi_1 \Delta^d \gamma_{c-1} + \cdots + \phi_p \Delta^d \gamma_{c-p} + \epsilon_c + \delta_1 \epsilon_{c-1} + \cdots + \delta_q \epsilon_{c-q}$$

Forecasting

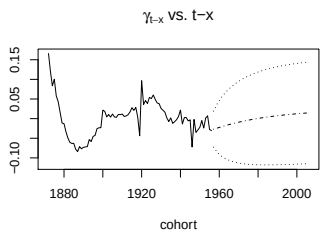
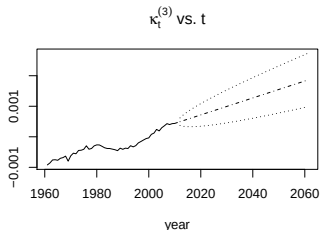
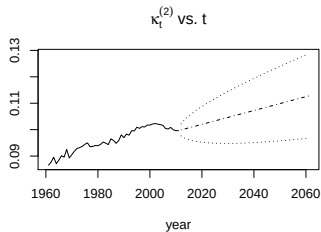
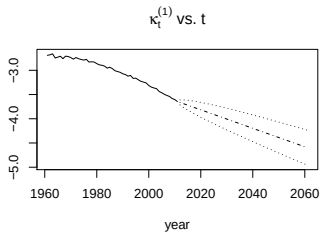
Model	Model for γ_{t-x}
APC	ARIMA(1, 1, 0) with drift
M7	ARIMA(2, 0, 0) with non-zero intercept

50-year ahead ($h = 50$) central projections: period indexes, cohort index, and one-year death probabilities:

```
LCfor <- forecast(LCfit, h=50)
CBDfor <- forecast(CBDfit, h=50)
APCfor <- forecast(APCfit, h=50, gc.order = c(1,1,0))
M7for <- forecast(M7fit, h=50, gc.order = c(2,0,0))
```

Forecasted period and cohort indexes

```
plot(M7for, parametricbx = FALSE)
```

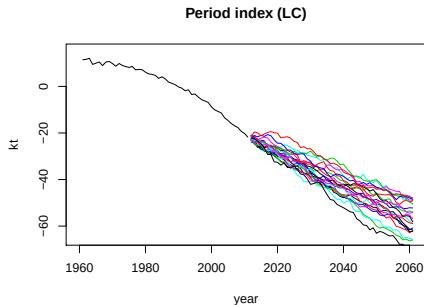


Simulation

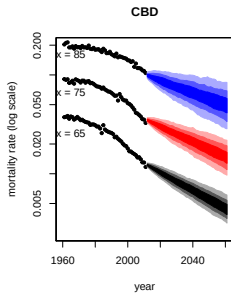
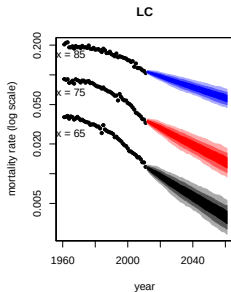
```
LCsim <- simulate(LCfit, nsim=500, h=50)
CBDsim <- simulate(CBDfit, nsim=500, h=50)
APCsim <- simulate(APCfit, nsim=500, h=50,
gc.order=c(1,1,0))
M7sim <- simulate(M7fit, nsim=500, h=50,
gc.order=c(2,0,0))
```

Simulation trajectories

```
#Plot period index trajectories for the LC model  
plot(LCfit$years, LCfit$kt[1,],  
xlim=c(1960,2061), ylim=c(-65,15),  
type="l", xlab="year", ylab="kt",  
main="Period index (LC)")  
matlines(LCsim$kt.s$years, LCsim$kt.s$sim[1,,1:20],  
type="l", lty=1)
```



Fancharts

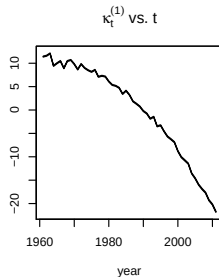
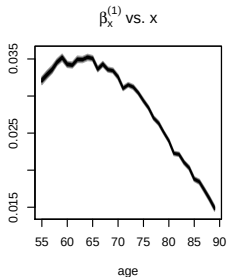
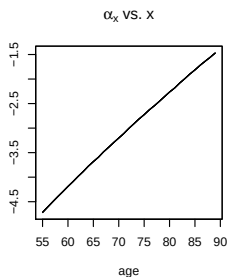


Parameter uncertainty and bootstrapping

StMoMo implements:

- Semiparametric bootstrapping (Brouhns et al., 2005)
- Residuals bootstrapping (Koissi et al., 2006)

```
LCboot <- bootstrap(LCfit, nBoot=500,  
type="semiparametric")  
plot(LCboot, nCol = 3)
```



Summary

- Use the framework of GLMs to define the GAPC family of models
- **StMoMo** uses this unifying framework to implement the vast majority of stochastic mortality models in the literature
 - ▷ Model fitting
 - ▷ Analysis of goodness-of-fit
 - ▷ Projection and simulations
 - ▷ Bootstrapping and parameter uncertainty
- Easy implementation and comparison of a wide range of models making it useful for:
 - ▷ Actuaries analysing longevity risk
 - ▷ Use in the classroom

References I

- Alai, D. H. and Sherris, M. (2014), “Rethinking age-period-cohort mortality trend models,” *Scandinavian Actuarial Journal*, 2014, 208–227.
- Aro, H. and Pennanen, T. (2011), “A user-friendly approach to stochastic mortality modelling,” *European Actuarial Journal*, 1, 151–167.
- Börger, M., Fleischer, D., and Kuksin, N. (2014), “Modeling the mortality trend under modern solvency regimes,” *ASTIN Bulletin: The Journal of the IAA*, 44, 1–38.
- Brouhns, N., Denuit*, M., and Van Keilegom, I. (2005), “Bootstrapping the Poisson log-bilinear model for mortality forecasting,” *Scandinavian Actuarial Journal*, 2005, 212–224.
- Butt, Z., Haberman, S., and Shang, H. (2014), “ilc: Lee-Carter mortality models using iterative fitting algorithms,” *URL* <http://cran.r-project.org/package=ilc>.

References II

- Cairns, A. J., Blake, D., and Dowd, K. (2006), “A two-factor model for stochastic mortality with parameter uncertainty: theory and calibration,” *Journal of Risk and Insurance*, 73, 687–718.
- Cairns, A. J., Blake, D., Dowd, K., Coughlan, G. D., Epstein, D., Ong, A., and Balevich, I. (2009), “A quantitative comparison of stochastic mortality models using data from England and Wales and the United States,” *North American Actuarial Journal*, 13, 1–35.
- Currie, I. D. (2016), “On fitting generalized linear and non-linear models of mortality,” *Scandinavian Actuarial Journal*, 2016, 356–383.
- Hunt, A. and Blake, D. (2018), “On the structure and classification of mortality models,” *North American Actuarial Journal*.

References III

- Hyndman, R. J., Booth, H., Tickle, L., and Maindonald, J. (2011), “demography: Forecasting mortality, fertility, migration and population data,” *R package version*, 1.
- Koissi, M.-C., Shapiro, A. F., and Högnäs, G. (2006), “Evaluating and extending the Lee–Carter model for mortality forecasting: Bootstrap confidence interval,” *Insurance: Mathematics and Economics*, 38, 1–20.
- Lee, R. D. and Carter, L. R. (1992), “Modeling and forecasting US mortality,” *Journal of the American statistical association*, 87, 659–671.
- O’Hare, C. and Li, Y. (2012), “Explaining young mortality,” *Insurance: Mathematics and Economics*, 50, 12–25.
- Plat, R. (2009), “On stochastic mortality modeling,” *Insurance: Mathematics and Economics*, 45, 393–404.

References IV

- Renshaw, A. E. and Haberman, S. (2003), “Lee–Carter mortality forecasting with age-specific enhancement,” *Insurance: Mathematics and Economics*, 33, 255–272.
- (2006), “A cohort-based extension to the Lee–Carter model for mortality reduction factors,” *Insurance: Mathematics and economics*, 38, 556–570.